Lecture 5

# Readings on intervals: spectral measures

**Objectives.** *By the spectral theorem, a cited result, self-adjoint operators correspond to spectral measures. We find the spectral measure explicitly for the position and the momentum and for every self-adjoint operator with an eigenbasis. The momentum with walls lies inside the integral of a spectral measure for every boundary condition, so a symmetric operator need not decide its readings.*

- Compute an interval projection and its probability.
- Apply the spectral-measure update after a yes.
- Explain why no state has both its position and its momentum in bounded intervals.

## 5.1 Spectral measures

Three rules for readings have appeared so far: the projections $E_a$ of an eigenbasis (Postulate 1.5), the projections $1_I$ of the position in a box (Postulate 2.6), and the projections $F^{-1}1_I F$ of the momentum on a line (Postulate 4.5). In each rule a projection belongs to a set of readings, and sets that exclude each other have orthogonal projections. This section collects what the three rules share.

In this lecture an *interval* is a set of real numbers of one of the forms $(a,b)$, $[a,b]$, $(a,b]$ or $[a,b)$, with $a \le b$. An endpoint may be $-\infty$ or $+\infty$ when its bracket is round. So $\mathbb{R}$ is an interval, and so are a single point $[a,a]$ and the empty set $(a,a)$.

**Definition 5.1** (Spectral measure)**.** A *spectral measure* on a Hilbert space $\mathcal{H}$ assigns to each interval $I$ an orthogonal projection $E(I)$ on $\mathcal{H}$ such that

1. $E(\emptyset) = 0$ and $E(\mathbb{R}) = \mathbb{1}$;
2. $E(I \cap J) = E(I)E(J)$ for all intervals $I$ and $J$;
3. $E(I \cup J) = E(I) + E(J)$ when $I$ and $J$ are disjoint and $I \cup J$ is an interval;
4. if $I_1 \subseteq I_2 \subseteq \cdots$ are intervals with the union $I$, then $E(I_n)\psi \to E(I)\psi$ for every $\psi \in \mathcal{H}$.

For disjoint intervals $I$ and $J$, normalisation and the intersection identity give $E(I)E(J) = E(\emptyset) = 0$. So the ranges of $E(I)$ and $E(J)$ are orthogonal, and readings that exclude each other have orthogonal projections, as in *Quantum Mechanics I*. For a vector $\psi \in \mathcal{H}$, put

$$\mu_\psi(I) = \langle \psi, E(I)\psi \rangle = \|E(I)\psi\|^2 . \tag{5.1}$$

For the box Hamiltonian, an interval selects the levels it contains. The interval $[3E_1, 10E_1]$ contains $E_2 = 4E_1$ and $E_3 = 9E_1$, so its projection is $|u_2\rangle\langle u_2| + |u_3\rangle\langle u_3|$. On $\psi = (u_1 + u_2)/\sqrt{2}$ its coefficients $c_n = \langle u_n, \psi\rangle$ give

$$\mu_\psi([3E_1, 10E_1]) = |c_2|^2 + |c_3|^2 = \tfrac{1}{2}.$$

For a unit vector $\psi$, normalisation gives $\mu_\psi(\mathbb{R}) = 1$. By additivity and the orthogonality, $\mu_\psi$ adds over two disjoint intervals, and by continuity on increasing intervals, $\mu_\psi(I_n) \to \mu_\psi(I)$. So $\mu_\psi$ is a probability distribution on the intervals, the *distribution* of $\psi$ under $E$.

A measure adds over countably many disjoint sets. The clauses ask for less, but they are enough to give such a measure, by a cited theorem. By the defining properties, the projections $E((-\infty, \lambda])$ form what Amrein, §4.2.1, calls a *spectral family*. For example, they are continuous from the right. Since the intervals $(\lambda + 1/n, \infty)$ increase to $(\lambda, \infty)$, normalisation, additivity and continuity on increasing intervals give, on every vector,

$$E((-\infty, \lambda + 1/n]) = \mathbb{1} - E((\lambda + 1/n, \infty)) \to \mathbb{1} - E((\lambda, \infty)) = E((-\infty, \lambda]).$$

By his (4.25)–(4.26), cited, the spectral measure extends to the *Borel sets*, the smallest family of sets of real numbers that contains every interval, the complement of each of its sets and the union of every sequence of its sets. By his (4.29) and continuity on increasing intervals the extension takes the value $E(I)$ on every interval. The extension adds over countably many disjoint sets $B_j$, in the sense that $E(\bigcup_j B_j)\psi = \sum_j E(B_j)\psi$ for every $\psi$. Integrals against $\mu_\psi$ are then defined. This lecture needs two kinds, and takes the following as their meaning. If $\mu_\psi(I) = \sum_{k:\,\lambda_k \in I} w_k$ for every interval $I$, for points $\lambda_k$ with weights $w_k \geq 0$, then

$$\int f \,\mathrm{d}\mu_\psi = \sum_k f(\lambda_k) w_k.$$

If $\mu_\psi(I) = \int_I w(\lambda)\,\mathrm{d}\lambda$ for every interval $I$, for a density $w$, then

$$\int f \,\mathrm{d}\mu_\psi = \int f(\lambda) w(\lambda)\,\mathrm{d}\lambda.$$

**Proposition 5.2** (Spectral measures from an eigenbasis and from indicators).

1. From an eigenbasis. *Let $(e_k)$ be an orthonormal basis of $\mathcal{H}$, with $k$ running through $\mathbb{N}$ or through $\mathbb{Z}$, and let the $\lambda_k$ be real. Then*

$$E(I)\psi = \sum_{k:\,\lambda_k \in I} \langle e_k, \psi\rangle \; e_k$$

   *defines a spectral measure.*
2. Indicators. *On $L^2(\mathbb{R})$, the projections $1_I$ of Definition 2.5, now on the line, form a spectral measure, and so do the projections $U^{-1}1_I U$, for every unitary $U$.*

*Proof.*

1. Write $c_k = \langle e_k, \psi\rangle$. Since $\sum_k |c_k|^2 = \|\psi\|^2$ by Parseval's identity, the series converges, and $E(I)$ is the orthogonal projection on the closed span of the $e_k$ with $\lambda_k \in I$. Normalisation holds, since $E(\emptyset)\psi$ is the empty sum and $E(\mathbb{R})\psi = \sum_k c_k e_k = \psi$ is the expansion of $\psi$ in the basis. The intersection identity and additivity hold because $E(I)$ keeps exactly the terms with $\lambda_k \in I$. For continuity on increasing intervals,

$$\|E(I)\psi - E(I_n)\psi\|^2 = \sum_{k:\,\lambda_k \in I \setminus I_n} |c_k|^2 .$$

Given $\epsilon > 0$, choose a finite set $K$ of indices with $\sum_{k\notin K} |c_k|^2 < \epsilon$. Each $\lambda_k \in I$ lies in some $I_n$, and then in all later ones. So for large $n$ no $k \in K$ has $\lambda_k \in I \setminus I_n$, and the sum is less than $\epsilon$.

2. As in § 2.4, each $1_I$ is an orthogonal projection. Normalisation, the intersection identity and additivity are the identities $1_\emptyset = 0$, $1_{\mathbb{R}} = 1$, $1_{I\cap J} = 1_I 1_J$ and $1_{I\cup J} = 1_I + 1_J$ for disjoint $I$ and $J$. For continuity on increasing intervals, let $\psi \in L^2(\mathbb{R})$ and $\epsilon > 0$. By the standing fact S4, there is a continuous $f$ that vanishes outside some $[-R, R]$, with $\|\psi - f\| < \epsilon$. Let $M$ bound $|f|$. If $I$ is empty there is nothing to prove. Otherwise let $a \le b$ be the endpoints of $I$ and, leaving out the empty $I_n$, let $a_n \le b_n$ be those of $I_n$. Then $a_n \to a$ and $b_n \to b$, and $I \setminus I_n$ lies in the closed intervals from $a$ to $a_n$ and from $b_n$ to $b$, which are half-lines when $a$ or $b$ is infinite. Let $\ell_n$ be the length of the part of these intervals inside $[-R, R]$. Since $a_n \to a$ and $b_n \to b$, the length $\ell_n$ tends to 0. Since $1_I - 1_{I_n} = 1_{I\setminus I_n}$, the triangle inequality gives the first step below. The projection $1_{I\setminus I_n}$ does not increase norms, and $|f| \le M$ vanishes outside $[-R, R]$, which gives the second:

$$\big\|1_I\psi - 1_{I_n}\psi\big\| \le \big\|1_{I\setminus I_n}(\psi - f)\big\| + \big\|1_{I\setminus I_n} f\big\| < \epsilon + M\sqrt{\ell_n},$$

which is less than $2\epsilon$ for large $n$. Finally, since $U^{-1} = U^\dagger$, each $U^{-1}1_I U$ is an orthogonal projection. Normalisation, the intersection identity and additivity carry over, since $U^{-1}1_I U\, U^{-1}1_J U = U^{-1}1_I 1_J U$ and conjugation by $U$ keeps sums, 0 and $\mathbb{1}$. Continuity on increasing intervals holds because $\big\|U^{-1}1_I U\psi - U^{-1}1_{I_n}U\psi\big\| = \big\|1_I U\psi - 1_{I_n}U\psi\big\|$. ∎

## 5.2 The spectral theorem

In *Quantum Mechanics I*, Postulate 4.1, an observable is a self-adjoint matrix $A = \sum_a a\, E_a$. For an operator with a domain, the sum becomes an integral, and the domain becomes a condition on the distributions. Let $E$ be a spectral measure and $A$ an operator. We write $A = \int \lambda \,\mathrm{d}E(\lambda)$ when

$$D(A) = \Big\{\psi : \int \lambda^2 \,\mathrm{d}\mu_\psi(\lambda) < \infty\Big\} \quad \text{and} \quad \langle\psi, A\psi\rangle = \int \lambda \,\mathrm{d}\mu_\psi(\lambda) \quad \text{for every } \psi \in D(A). \tag{5.2}$$

Since $|\lambda| \le 1 + \lambda^2$ and $\mu_\psi(\mathbb{R}) = \|\psi\|^2$ is finite, the second integral converges when the first does. For a unit vector $\psi$, the first condition says that the distribution $\mu_\psi$ has a finite mean square, and the second that $\langle\psi, A\psi\rangle$ is its mean.

> **Theorem 5.3** (The spectral theorem · *cited*)**.** *For every spectral measure $E$ there is exactly one operator $A$ with $A = \int \lambda \,\mathrm{d}E(\lambda)$. It is self-adjoint, and $\|A\psi\|^2 = \int \lambda^2 \,\mathrm{d}\mu_\psi(\lambda)$ for every $\psi \in D(A)$. Conversely, for every self-adjoint operator $A$ there is exactly one spectral measure $E_A$ with $A = \int \lambda \,\mathrm{d}E_A(\lambda)$.*

So $E \mapsto \int \lambda \,\mathrm{d}E$ matches the spectral measures one to one with the self-adjoint operators. The first part is Hall, Props 10.1–10.3: the means $\langle\psi, A\psi\rangle$ on a dense domain fix the operator $A$. The converse is Hall, Thm 10.4, whose measures live on the Borel subsets of a set that Lecture 6 calls the spectrum, extended by zero. Amrein, Prop. 4.30, gives the uniqueness among all spectral families $E_A((-\infty, \lambda])$. By the domain in (5.2), a unit vector $\psi$ lies in $D(A)$ exactly when its distribution $\mu_\psi$ under $E_A$ has a finite mean square, and by the formula for $\|A\psi\|^2$ this mean square is $\|A\psi\|^2$, as for the box's energy in § 1.7.

**Proposition 5.4** (Explicit spectral measures)**.**

1. An eigenbasis. *Let $(e_k)$ and $\lambda_k$ be as in Proposition 5.2, with the spectral measure $E$ made from them, and write $c_k = \langle e_k, \psi\rangle$. The operator $B$ with*

$$D(B) = \left\{\psi : \sum_k \lambda_k^2 |c_k|^2 < \infty\right\}, \qquad B\psi = \sum_k \lambda_k c_k e_k,$$

*is self-adjoint, and $B = \int \lambda \, \mathrm{d}E(\lambda)$. If a self-adjoint operator $A$ has $Ae_k = \lambda_k e_k$ for every $k$, then $A = B$.*

2. Position and momentum. *On $L^2(\mathbb{R})$, $Q = \int \lambda \, \mathrm{d}E_Q(\lambda)$ with $E_Q(I) = 1_I$, and $P = \int \lambda \, \mathrm{d}E_P(\lambda)$ with $E_P(I) = F^{-1}1_I F$.*

*Proof.*

1. The steps of the proof of Theorem 1.8 on density, symmetry and the adjoint apply word for word, with $e_k$ and $\lambda_k$ in place of $u_n$ and $E_n$, so $B$ is self-adjoint. Next, $\mu_\psi(I) = \sum_{k:\,\lambda_k \in I} |c_k|^2$ has the points $\lambda_k$ with the weights $|c_k|^2$. So $\int \lambda^2 \, \mathrm{d}\mu_\psi = \sum_k \lambda_k^2 |c_k|^2$, and by Parseval's identity $\langle \psi, B\psi\rangle = \sum_k \lambda_k |c_k|^2 = \int \lambda \, \mathrm{d}\mu_\psi$. Finally, let $A$ be self-adjoint with $Ae_k = \lambda_k e_k$, and let $\psi \in D(A)$. Since $\lambda_k$ is real, the symmetry of $A$ gives $\langle e_k, A\psi\rangle = \langle Ae_k, \psi\rangle = \lambda_k c_k$. By Parseval's identity, $\sum_k \lambda_k^2 |c_k|^2 = \|A\psi\|^2 < \infty$, so $\psi \in D(B)$ and $A\psi = B\psi$. Hence $A \subseteq B$, and since adjoints reverse inclusions (Proposition 0.5), $B = B^\dagger \subseteq A^\dagger = A$. So $A = B$.
2. By Proposition 5.2, $E_Q$ and $E_P$ are spectral measures. For $E_Q$, the distribution $\mu_\psi(I) = \int_I |\psi(x)|^2 \, \mathrm{d}x$ has the density $|\psi|^2$. So $\int \lambda^2 \, \mathrm{d}\mu_\psi = \int x^2 |\psi(x)|^2 \, \mathrm{d}x$, which is finite exactly when $\psi \in D(Q)$, and $\langle \psi, Q\psi\rangle = \int x |\psi(x)|^2 \, \mathrm{d}x = \int \lambda \, \mathrm{d}\mu_\psi$. For $E_P$, since $F$ is unitary, $\mu_\psi(I) = \|1_I F\psi\|^2 = \int_I \left|\tilde{\psi}(p)\right|^2 \mathrm{d}p$ has the density $\left|\tilde{\psi}\right|^2$, and the same computation with $p$ and $\tilde{\psi}$ gives $P$ (Definition 4.4). ∎

By the uniqueness in Theorem 5.3, $E$ is the spectral measure of $B$, and $E_Q$ and $E_P$ are those of $Q$ and $P$. On the Borel sets $B$, the extension of $E_Q$ is multiplication by the indicator of $B$ (Amrein, Example 4.9). For the box Hamiltonian, Theorem 1.8 gives the eigenbasis $u_n$, so

$$E_H(I)\psi = \sum_{n:\,E_n \in I} c_n u_n, \qquad c_n = \langle u_n, \psi\rangle,$$

and in particular $E_H(\{E_n\}) = |u_n\rangle\langle u_n|$. The spectral measure is made from an eigenbasis in the same way for the circle's momentum, with the eigenbasis $f_k$ of Proposition 3.2, and for each $P_\theta$, with the eigenbasis $e_{\theta,k}$ of Theorem 3.10, whose proof is sketched.

**Self-check 5.2**

1. For which numbers $\lambda$ is $E_{P_\pi}(\{\lambda\}) \neq 0$? Which projection is $E_{P_0}(\{0\})$?

## 5.3 Reading an interval

**Postulate 5.5** (Readings on intervals)**.** The observables of a quantum system are self-adjoint operators. Let $A$ be an observable and $\psi$ a unit vector. An ideal reading of "$A$ lies in $I$", for an interval $I$, on the state of $\psi$ answers yes with the probability

$$\mathrm{Prob}(A \in I) = \|E_A(I)\psi\|^2 . \tag{5.3}$$

An answer that occurs has positive probability. If it is yes, the state immediately

afterwards is the ray of $E_A(I)\psi/\|E_A(I)\psi\|$. If it is no, the state is the ray of $(\mathbb{1} - E_A(I))\psi/\|(\mathbb{1} - E_A(I))\psi\|$.

This is Born's and Lüders' rule of *Quantum Mechanics I*, Postulate 4.1, for the projection $E_A(I)$. Amrein, p. 168, states its probability in the same form. The postulate contains the three earlier rules. For an observable with an eigenbasis, $E_A(\{a\})$ is the projection $E_a$ of Postulate 1.5. On the line, Proposition 5.4 gives the projections $1_I$ and $F^{-1}1_I F$ of Postulate 4.5. In the box, the same proof on $L^2(0, L)$ gives $E_Q(I) = 1_{I\cap[0,L]}$ for the position $Q$ of § 4.4. For $I \subseteq [0, L]$ this is the projection $1_I$ of Postulate 2.6.

In the state immediately after a yes on $I$, the distribution of $A$ is its earlier distribution conditioned on $I$. Let $\varphi = E_A(I)\psi/\|E_A(I)\psi\|$. By the intersection identity of Definition 5.1, $E_A(J)E_A(I) = E_A(I \cap J)$, so for every interval $J$

$$\mathrm{Prob}_\varphi(A \in J) = \frac{\|E_A(I \cap J)\psi\|^2}{\|E_A(I)\psi\|^2} = \frac{\mathrm{Prob}_\psi(A \in I \cap J)}{\mathrm{Prob}_\psi(A \in I)}.$$

This is Bayes' rule for the event that $A$ lies in $I$. A reading of another observable need not follow such a rule, as the example below shows.

A single point is an interval. Multiplication by the indicator of $\{x\}$ maps every $\psi$ to a function that vanishes off the point $x$, which is the zero class. So $E_Q(\{x\}) = 0$. A position reading at a point has the probability 0 in every state, so its yes never occurs, and no state follows it. For the box's energy, by contrast, $E_H(\{E_n\}) = |u_n\rangle\langle u_n| \neq 0$.

Physics texts write $E_Q(I) = \int_I |x\rangle\langle x|\,\mathrm{d}x$. There is no vector $|x\rangle$ in $L^2(\mathbb{R})$, since a function that vanishes off the point $x$ is the zero class. So $|x\rangle\langle x|\,\mathrm{d}x$ is notation for $\mathrm{d}E_Q(x)$, used only under an integral, where it stands for $\langle\varphi, E_Q(I)\psi\rangle = \int_I \overline{\varphi(x)}\,\psi(x)\,\mathrm{d}x$.

Figure 5.1 shows the window reading and the change in momentum.

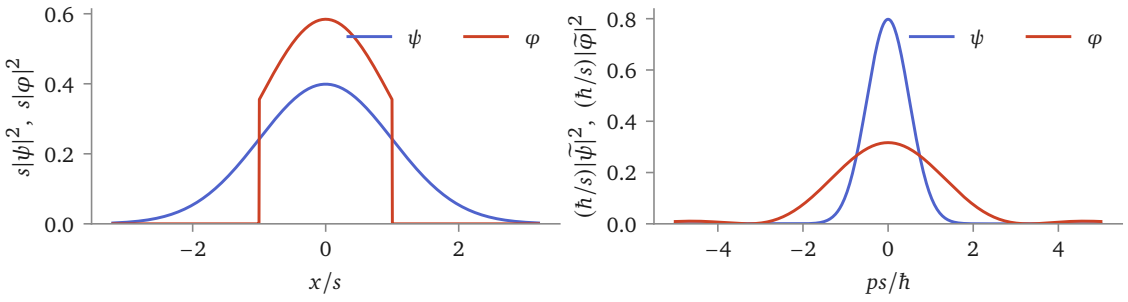



Figure 5.1: The initial Gaussian $\psi$ and the state $\varphi$ after a yes on $[-s, s]$. The cut in the position density produces broad momentum tails.

**Example 5.6** (A Gaussian read through a window)**.** Take the packet of Theorem 4.9 with $q = p_0 = 0$,

$$\psi(x) = (2\pi s^2)^{-1/4} e^{-x^2/4s^2},$$

and the window $I = [-s, s]$.

1. *The reading.* By (5.3) and Proposition 5.4, the position reads in $I$ with the probability $c_I = \int_{-s}^{s} |\psi(x)|^2\,\mathrm{d}x \approx 0.683$, the probability that a normal reading lies within one spread of its mean.
2. *The state after a yes.* By the postulate, it is the ray of $\varphi = 1_I\psi/\sqrt{c_I}$, the packet cut off at $\pm s$. At $\pm s$ the function $\varphi$ jumps to 0 from $\varphi(s) = (2\pi s^2)^{-1/4} e^{-1/4}/\sqrt{c_I}$. By the conditioning rule, the position density of $\varphi$ is $|\psi|^2/c_I$ on $I$ and 0 outside.

3. *The momentum before.* By Example 4.12, the momentum density of $\psi$ is normal, with the mean 0 and the spread $\hbar/2s$.
4. *The momentum after.* Since $\varphi$ vanishes outside $I$, its transform is the integral (4.1). This formula is part of Plancherel's theorem, the standing fact S1. Indeed, Hall, Thm A.19, computes $F\varphi$ as the limit of the integrals over $[-R, R]$, and for $\varphi$ these do not change once $R \geq s$. So

$$\tilde{\varphi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-s}^{s} e^{-\mathrm{i}px/\hbar} \varphi(x)\, \mathrm{d}x,$$

an error function of a complex argument, computed numerically. A momentum reading lies in $[-a, a]$ with the probabilities

| $a$ | $\hbar/2s$ | $\hbar/s$ | $2\hbar/s$ | $10\hbar/s$ |
|---|---|---|---|---|
| before, $\psi$ | 0.683 | 0.954 | 1.000 | 1.000 |
| after, $\varphi$ | 0.309 | 0.572 | 0.872 | 0.976 |

The reading of the window has spread the momentum readings far beyond $\hbar/2s$.

5. *An infinite spread.* Because $\varphi$ jumps at $\pm s$, two integrations by parts give, for $p \neq 0$,

$$\tilde{\varphi}(p) = \left(\frac{2\hbar}{\pi}\right)^{1/2} \varphi(s)\, \frac{\sin(ps/\hbar)}{p} + r(p), \qquad |r(p)| \leq \frac{C}{p^2},$$

for a constant $C$ (Exercise Sheet 5). So for large $|p|$,

$$p^2\, |\tilde{\varphi}(p)|^2 \approx \frac{2\hbar}{\pi}\, \varphi(s)^2 \sin^2 \frac{ps}{\hbar},$$

whose average over a period is half its maximum. So $\int p^2\, |\tilde{\varphi}(p)|^2\, \mathrm{d}p = \infty$. Hence $\varphi \notin D(P)$, and $\Delta P = \infty$ in the convention of § 2.6. Even $\int |p|\, |\tilde{\varphi}(p)|^2\, \mathrm{d}p$ diverges (Exercise Sheet 5), so the momentum readings of $\varphi$ have no mean.

This is the line's form of the reading of § 2.6. There a sharp reading of position left an infinite spread of the energy, and here it leaves an infinite spread of the momentum. The momentum readings are not conditioned on the window, as the position readings are. The reading has changed them.

**Self-check 5.3**

1. For $\psi$ of Example 5.6, a student writes the probability that the position reads in $[0, s]$ as $\int_0^s x\, |\psi(x)|^2\, \mathrm{d}x$. What does this integral compute, and what is the probability?

## 5.4 A formula need not decide the readings

Self-adjointness determines the rule for readings. Theorem 5.3 gives a first answer. A rule for the readings of an observable is a spectral measure, and by the theorem the spectral measures match the self-adjoint operators. The momentum with walls $P_D$ of Definition 3.5 is symmetric and not self-adjoint (Proposition 3.7). So no spectral measure has $P_D$ as its integral. Instead, many such integrals extend $P_D$ in the sense of Definition 0.4, and we say that $P_D$ *lies inside* each of them.

**Proposition 5.7** (The momentum with walls lies inside every $\int \lambda\, \mathrm{d}E_\theta$). *For $\theta \in [0, 2\pi)$, let $\lambda_{\theta,k} = \hbar(\theta + 2\pi k)/L$, and let $E_\theta$ be the spectral measure made from the orthonormal basis*

*$e_{\theta,k}$ of Theorem 3.10 and the numbers $\lambda_{\theta,k}$, as in Proposition 5.2. Let $B_\theta = \int \lambda \, \mathrm{d}E_\theta(\lambda)$ be the operator of Proposition 5.4. Then $P_D \subseteq B_\theta$ for every $\theta$, and $E_\theta \neq E_{\theta'}$ when $\theta \neq \theta'$.*

*Proof.* As in the proof of Theorem 3.10, $e_{\theta,k} = e^{\mathrm{i}\theta x/L} f_k$, so by the standing fact S2 the $e_{\theta,k}$ form an orthonormal basis. Let $\psi \in D(P_D)$, and write $c_k = \langle e_{\theta,k}, \psi \rangle$. The function $e_{\theta,k}$ lies in $\mathcal{A}$, with $-\mathrm{i}\hbar e'_{\theta,k} = \lambda_{\theta,k} e_{\theta,k}$, and $\psi(0) = \psi(L) = 0$. So the boundary form (3.2) gives

$$\langle e_{\theta,k}, P_D\psi \rangle = \langle -\mathrm{i}\hbar e'_{\theta,k}, \psi \rangle = \lambda_{\theta,k} c_k,$$

since $\lambda_{\theta,k}$ is real. By Parseval's identity, $\sum_k \lambda_{\theta,k}^2 |c_k|^2 = \|P_D\psi\|^2 < \infty$, so $\psi \in D(B_\theta)$ and $B_\theta\psi = \sum_k \langle e_{\theta,k}, P_D\psi \rangle \, e_{\theta,k} = P_D\psi$. Finally, $E_\theta(\{\lambda\}) \neq 0$ exactly when $\lambda$ is one of the $\lambda_{\theta,k}$. For $\theta \neq \theta'$ in $[0, 2\pi)$ the sets $\hbar(\theta + 2\pi\mathbb{Z})/L$ and $\hbar(\theta' + 2\pi\mathbb{Z})/L$ are disjoint, since $\theta - \theta'$ is not a multiple of $2\pi$. So $E_\theta(\{\lambda_{\theta,0}\}) \neq 0 = E_{\theta'}(\{\lambda_{\theta,0}\})$. ∎

By Proposition 5.4, $B_\theta$ is $P_\theta$, and $E_\theta$ is $E_{P_\theta}$. For the ground state $u_1 \in D(P_D)$, Example 3.12 found that $E_0$ and $E_\pi$ give different distributions with the same mean and the same mean square. So the rule $-\mathrm{i}\hbar\psi'$ with the walls fixes the mean and the mean square of the readings on $D(P_D)$, but not the readings themselves. By Proposition 3.11, whose proof is sketched, and Theorem 5.3, the $E_\theta$ are all the spectral measures whose integral extends $P_D$. To read the momentum of a particle in a box, one must choose $\theta$. A symmetric operator need not decide the probabilities of its readings, and a self-adjoint operator does. Lecture 7 gives a second answer, from evolution.

**Self-check 5.4**

1. For which $\theta$ is $E_\theta(\{0\}) \neq 0$? What is then the probability that the reading is 0 for $u_1$?

## 5.5 Position and momentum cannot both be bounded

A state cannot have both its position and its momentum readings in bounded intervals. In *Quantum Mechanics I*, Thm 9.3, a nonzero vector of $\mathbb{C}^N$ and its discrete Fourier transform cannot both have few nonzero entries. On the line the answer is sharper.

**Theorem 5.8** (Position and momentum cannot both be bounded). *Let $\psi \in L^2(\mathbb{R})$ be nonzero and vanish outside a bounded interval. Then $\tilde{\psi}$ vanishes on no interval of positive length. In particular, $E_P(J)\psi \neq 0$ for every interval $J$ of positive length, and no nonzero $\psi$ has both $E_Q(I)\psi = \psi$ and $E_P(J)\psi = \psi$ for bounded intervals $I$ and $J$.*

*Proof.* Choose $X$ with $\psi = 0$ outside $[-X, X]$. As in Example 5.6, the transform is the integral

$$\tilde{\psi}(p) = \frac{1}{\sqrt{2\pi\hbar}} \int_{-X}^{X} e^{-\mathrm{i}px/\hbar} \psi(x) \, \mathrm{d}x \quad \text{for every } p.$$

Fix $p_0$. For $|x| \leq X$, the exponential series

$$e^{-\mathrm{i}px/\hbar} = e^{-\mathrm{i}p_0 x/\hbar} \sum_{n=0}^{\infty} \frac{1}{n!} \left( \frac{-\mathrm{i}(p - p_0)x}{\hbar} \right)^n$$

has terms of modulus at most $(|p - p_0| X/\hbar)^n / n!$. So, for every such $x$, its remainder after $N$ terms is at most the remainder $\epsilon_N$ of the series of $e^{|p-p_0|X/\hbar}$. By Cauchy–Schwarz on $[-X, X]$,

$\int_{-X}^{X} |\psi| \le \sqrt{2X}\ \|\psi\|$. So the integral of the remainder against $\psi$ is at most $\epsilon_N \sqrt{2X}\ \|\psi\|$, which tends to 0, and

$$\tilde{\psi}(p) = \sum_{n=0}^{\infty} a_n (p - p_0)^n, \qquad a_n = \frac{1}{\sqrt{2\pi\hbar}} \frac{1}{n!} \Big(\frac{-\mathrm{i}}{\hbar}\Big)^n \int_{-X}^{X} x^n e^{-\mathrm{i}p_0 x/\hbar} \psi(x)\, \mathrm{d}x,$$

for every real $p$. So $\tilde{\psi}$ is a power series about every point, and it converges on the whole line. Now suppose that $\tilde{\psi}$ vanishes on an interval $J$ of positive length. As a power series, $\tilde{\psi}$ is continuous, and by the first fact about $\mathcal{A}$ in § 3.2, a continuous function in the zero class on $J$ vanishes at every point of $J$. Take $p_0$ inside $J$. Since $\tilde{\psi}$ vanishes on an interval around $p_0$, every derivative of $\tilde{\psi}$ at $p_0$ is 0. The $n$th derivative of the series at $p_0$ is $n!\, a_n$, so every $a_n$ is 0. Since this series converges on the whole line, $\tilde{\psi} = 0$ there, and $\psi = 0$ by the standing fact S1, a contradiction.

The last two claims follow. First, $\|E_P(J)\psi\|^2 = \int_J |\tilde{\psi}|^2$, which is positive. Next, let a nonzero $\psi$ have $E_Q(I)\psi = \psi$ and $E_P(J)\psi = \psi$ for bounded intervals $I$ and $J$. Then $\psi$ vanishes outside $I$, so $\tilde{\psi}$ vanishes on no interval of positive length. But $E_P(J)\psi = \psi$ says that $\tilde{\psi}$ vanishes outside $J$, and so on a half-line, a contradiction. ∎

So the state $\varphi$ that the window of Example 5.6 leaves gives every interval of momenta of positive length a positive probability. A state can have its position readings in a bounded interval, or its momentum readings, but not both. The Gaussian packets have neither, since their densities are positive everywhere. The theorem concerns the intervals that hold the readings, while Theorem 4.9 concerns their spreads.

**Self-check 5.5**

1. Let $\psi$ vanish outside $[-X, X]$. Can $\tilde{\psi}$ vanish at a single point? Try $\psi = 1_{[-X,X]}/\sqrt{2X}$.

**The physics**

In 2009 Bakr, Gillen, Peng, Fölling and Greiner imaged single atoms of rubidium-87 held in an optical lattice, a pattern of laser light whose potential minima form a square grid. The grid had the period 640 nm, and their microscope resolved about 600 nm, so each image showed which sites held an atom. In 2010 Sherson, Weitenberg, Endres, Cheneau, Bloch and Kuhr did the same in a lattice of period 532 nm. For a single atom, the question "is the atom on this site?" is a reading of its position on a square cell of the lattice, the two-dimensional form of a reading on an interval. Atoms on one site are lost in pairs during the imaging, so each image shows whether the number of atoms on a site was odd or even.

**Readings on intervals**

*Observables:* by the cited spectral theorem, every self-adjoint $A$ is $\int \lambda\, \mathrm{d}E_A$ for exactly one spectral measure $E_A$, and $D(A)$ holds the vectors of finite mean square (Theorem 5.3). For $Q$, $P$ and every self-adjoint operator with an eigenbasis, $E_A$ is explicit (Proposition 5.4). The symmetric $P_D$ lies inside the integral of every $E_\theta$, so it does not decide its readings (Proposition 5.7). *Measurement:* a reading of "$A$ lies in $I$" on a unit vector $\psi$ answers yes with probability $\|E_A(I)\psi\|^2$, and a yes leaves the ray of $E_A(I)\psi/\|E_A(I)\psi\|$ (Postulate 5.5). A Gaussian read through a window leaves $\Delta P = \infty$ (Example 5.6). No nonzero state has both its position and its momentum in bounded intervals (Theorem 5.8).

**Summary.**

- *Spectral measures*: projections on intervals with the defining properties, extended to the Borel sets by a cited theorem (Definition 5.1). An eigenbasis, the indicators on the line and their unitary images give spectral measures (Proposition 5.2).
- *The spectral theorem*: the spectral measures match the self-adjoint operators one to one, with $D(A)$ the vectors of finite mean square (Theorem 5.3, cited). The spectral measure is explicit for $Q$, $P$ and every self-adjoint operator with an eigenbasis (Proposition 5.4).
- *Readings on intervals*: probability $\|E_A(I)\psi\|^2$, and a yes leaves $E_A(I)\psi/\|E_A(I)\psi\|$ (Postulate 5.5). A Gaussian read through a window leaves $\Delta P = \infty$ (Example 5.6).
- *The momentum with walls lies inside the integral of every* $E_\theta$, so a symmetric operator need not decide its readings (Proposition 5.7).
- *Position and momentum cannot both be bounded*: a nonzero state that vanishes outside a bounded interval has a transform that vanishes on no interval of positive length (Theorem 5.8).

Lecture 6 locates the readings of an observable without a diagonalisation, in its spectrum.

**Self-check answers.**

**5.2** 1. Exactly for the eigenvalues $(2k+1)\pi\hbar/L$, $k \in \mathbb{Z}$, of $P_\pi$. The projection $E_{P_0}(\{0\})$ is $|f_0\rangle\langle f_0|$, on the constant functions.

**5.3** 1. It is the part of the mean position $\int x\,|\psi|^2\,\mathrm{d}x$ that comes from $[0,s]$, a length, not a probability: $s(1-e^{-1/2})/\sqrt{2\pi} \approx 0.157\,s$. The probability is $\int_0^s |\psi|^2\,\mathrm{d}x = c_I/2 \approx 0.341$.

**5.4** 1. Only for $\theta = 0$, where $0 = \lambda_{0,0}$. Then $E_0(\{0\}) = |f_0\rangle\langle f_0|$, and the probability is $|\langle f_0, u_1\rangle|^2 = 8/\pi^2 \approx 0.811$.

**5.5** 1. Yes. For $\psi = 1_{[-X,X]}/\sqrt{2X}$, $\tilde\psi(p) = (\hbar/\pi X)^{1/2}\sin(pX/\hbar)/p$, which vanishes at $p = \pi\hbar/X$. A single point is not an interval of positive length.

## References

Hall, *Quantum Theory for Mathematicians*, Props 10.1–10.3 (pp. 202–204), Thm 10.4 (p. 205), Thm A.19 (p. 532). Amrein, *Hilbert Space Methods in Quantum Mechanics*, §4.2.1 (pp. 141–143), Example 4.9 (p. 144), p. 168, Prop. 4.30 (p. 173). *Quantum Mechanics I*, Postulate 4.1 and Thm 9.3. Bakr, Gillen, Peng, Fölling and Greiner, *Nature* 462, 74 (2009). Sherson, Weitenberg, Endres, Cheneau, Bloch and Kuhr, *Nature* 467, 68 (2010).